INCH-POUND

MIL-T-43819B
26 November 1991
SUPERSEDING
MIL-T-43819A
30 March 1984

MILITARY SPECIFICATION

TRAY, MESS, COMPARTMENTED, PAPER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a disposable, paper, 5-compartment mess tray (see 6.1).

1.2 Classification. The 5-compartment mess trays shall be of one type in the following classes (see 6.2):

Class 1 - White or natural
Class 2 - Green

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 7350

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are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-1175 - Tray, Mess, 5-Compartment, Paper

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UX, Natick, MA 01760-5017.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 170-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-9325.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

TECHNICAL ASSOCIATION OF THE PULP AND PAPER INDUSTRY (TAPPI)

Standard T402-OM-88 - Conditioning Paper and Paperboard for Testing

(Applications for copies should be addressed to the Technical Association of the Pulp and Paper Industry, P.O. Box 105113, Atlanta, GA 30348.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard product. The trays shall be the manufacturer's standard commercial product, except for any changes necessary to meet document requirements.

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3.3 Material. Pulp for the trays shall be produced from chemical or mechanical pulp. Additives of after-treatments may be used provided the materials, processes, and finished tray comply with all requirements of this document. The molded pulp for the trays and the anti-oxidants or other substances incorporated in the molded pulp shall meet the requirements of the Federal Food, Drug, and Cosmetic Act as amended: Title 21, CFR 176.170 (see 4.4.1.1). It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.4 Design and construction. The compartmented tray shall conform to Drawing 5-11-1175. The compartmented tray shall be molded construction and the trays shall be nestable.

3.4.1 Rigidity. The average force needed to deflect the trays a distance of 1 cm shall be no less than 450 grams in the length direction and no less than 500 grams in the width direction when tested as specified in 4.5.1.2.

3.4.2 Water resistance. Trays shall not show total penetration of water, ply separation, or delamination more than 1/4 inch in any dimension, or any blister more than 1/4 inch in diameter, when in contact with water for 30 minutes and at a water temperature between 160°F and 162°F when tested as specified in 4.5.2. Total penetration is defined as the wetting and staining of the bond paper under the tray.

3.4.3 Odor and taste. Trays shall not have any objectionable odor and shall not impart any objectionable taste to water when tested as specified in 4.5.3.

3.4.4 Grease resistance. Trays shall be grease resistant and shall not show evidence of total penetration by oil when tested as specified in 4.5.4. Total penetration is defined as the wetting and staining of the bond paper under the tray.

3.5 Finish. Trays shall have a hot die dried pressed finish. The color of class 1 trays shall be white or natural pulp. The color of class 2 trays shall approximate any of the following colors of FED-STD-595: 34373, 34441, 34491, 34504, 34516, 34518, 34558, or 34670.

3.6 Marking for identification. The trays shall be marked with the manufacturer's name, trade name, or trademark readily identifiable with the manufacturer.

3.7 Workmanship. All paper stock for the trays shall be protected from contamination during and after processing and shall be clean and free from holes, tears, cuts, cracks, scuffs, buckled areas, delamination, blister, distortion, malformation, or warpage. The end item shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in table I and 4.4.3 and tested for the characteristics specified in table II. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents, unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Component and materials certification. A certificate of compliance may be acceptable as evidence that the molded pulp, anti-oxidants, or other substances incorporated in the molded pulp for the trays conforms to the Title 21, CFR requirements of 3.3.

4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of one tray. The sample unit shall be one completely fabricated tray. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0 for major defects and 6.5 for total (major and minor combined) defects.

TABLE I. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Not as specified	101	
	Not color specified		201
	Rough edges	102	
	Not free of dust, dirt, grease, oil, or foreign matter imbedded in molded pulp	103	
Construction and workmanship	Not design specified	104	
	Number of compartments not as specified	105	
	Trays do not nest	106	
	Not clean and free from holes, tears, cuts, cracks, scuffed, or buckled areas	107	
	Any delamination, blister, distortion, malformation, or warpage	108	
Marking	Missing, not legible, incomplete, or not as specified		202

4.4.3 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified on the drawing. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of one box of trays (see 5.2.1 and 5.2.2). The sample unit shall be one completely fabricated tray. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units shall be 2.5.

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4.4.4 End item testing. The end item shall be tested for the characteristics specified in table II. The lot size shall be expressed in units of trays. The sample unit shall be one tray. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method
Rigidity	3.4.1	4.5.1
Water resistance	3.4.2	4.5.2
Odor and taste	3.4.3	4.5.3
Grease resistance	3.4.4	4.5.4

4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be

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expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified Interlocking of loads not as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 Rigidity test.

4.5.1.1 Apparatus. Single Service Institute Rigidity Tester with Ametek Load Cell, Model No. ML4431-2 or equivalent (see 6.4).

4.5.1.2 Procedure. All samples are to be conditioned in accordance with TAPPI method T402-OM-88. Load cell is to be set up and test performed in accordance with instructions furnished with the testing device. Inability of the tray to meet the specified requirements for rigidity in 3.4.1 shall constitute failure of this test.

4.5.2 Water resistance test. Distilled water at a temperature between 160°F and 162°F shall be poured onto the tray surfaces to a depth approximately equal to one-half the depth of the tray. The tray shall rest on a flat sheet of double face corrugated board or other suitable paper support, approximately 1/4 inch thick, with a sheet of clean white bond paper between the tray and supporting surface. After 30 minutes the water in the test tray shall be drained into a flask and retained for use in the odor and taste test. The test tray shall then be examined for evidence of delamination. Separation of plies extending more than 1/4 inch in any direction or formation of blisters greater than 1/4 inch in diameter shall be considered delamination. The bond paper and underside of the tray shall be examined for visible evidence of water leakage to determine compliance with 3.4.2. Slight dampness of the paper due to condensation shall not be construed as a failure. Inability to meet the requirements for water resistance specified shall constitute failure of this test.

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4.5.3 Odor and taste test. The distilled water used in the water resistance test (see 4.5.2) shall be collected and allowed to stand in a stoppered flask or bottle for 2 hours. The odor and taste of the sample water shall be compared with that of freshly distilled water at the same temperature to determine compliance with 3.4.3. Inability to meet the requirements for odor and taste specified shall constitute failure of this test.

4.5.4 Grease resistance test. Place a mixture of 3.0 milliliters of corn oil, of the grade used for cooking, and 10 grams of clean, dry sand, of such a particle size that all of it will pass through a No. 20 U.S. Standard sieve and be retained by a No. 30 U.S. Standard sieve, in a glass beaker of approximately 30 milliliter capacity and heat in an oven to 160°F ± 2°F. Remove the hot mixture from the oven and swirl the beaker contents to assure good mixing of the oil and sand. Cover the beaker with the tray to be tested, using the eating surface of the tray as the cover. Invert the tray and beaker and place the tray on a hard, flat surface with a sheet of clean white bond paper directly under the tray, with the beaker in an inverted position over the sand and oil mixture. Allow the mixture to stand, as specified above, for 30 minutes. At the end of this test period, examine the bond paper and the bottom of the tray for any evidence of staining, to determine compliance with 3.4.4. Inability to meet the requirements for grease resistance specified shall constitute failure of this test.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. Eighteen trays shall be nested and unit packed in a minimum 0.001 inch thick snug-fitting polyethylene bag. The bag shall be closed in a manner which will prevent accidental opening.

5.2 Packing. Packing shall be Commercial.

5.2.1 Commercial packing. Five hundred and four trays, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Unit loading. Trays, packed as specified, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Marking.

5.4.1 Unit packs and shipping containers. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

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5.4.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Compartmented paper trays are intended for one-time use in military feeding systems and all systems using the Tray Pack Ration.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample consisting of three trays. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Rigidity tester. The apparatus referred to in 4.5.1.1 may be obtained from the Foodservice and Packaging Institute, 1025 Connecticut Avenue, N.W., Washington, DC 20036.

6.5 Subject term (key word) listing.

Disposable
Feeding unit, divided
Molded pulp

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Review activities:

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Army - GL

Project 7350-0252

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-43819B	2. DOCUMENT DATE (YYMMDD) 91 November 26

3. DOCUMENT TITLE: TRAY, MESS, COMPARTMENTED, PAPER

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY		
a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial 508-651-4085	(2) AUTOVON 256-4085
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-WEE	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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